

Sanraksha App

ABSTRACT

Sanraksha Alert is an Android-based personal emergency safety application designed to provide rapid SOS assistance in life-threatening situations. Built entirely as a **local-first system** with no server dependency, the app empowers users to send distress signals through multiple intelligent triggers and communication channels.

The application supports **four SOS activation mechanisms** — a manual button with a 30-second countdown, accelerometer-based shake detection, continuous voice recognition listening for keywords ("help"/"SOS"), and ambient sound detection for scream-level audio. Upon activation, the app simultaneously plays a siren alarm, retrieves the user's real-time GPS coordinates, dispatches emergency SMS alerts with a Google Maps location link to all registered emergency contacts, and auto-dials the primary contact.

A distinctive **Kidnapping Mode** enables sustained location tracking, broadcasting the user's GPS trail every 5 minutes as a continuous Google Maps direction path via SMS — allowing rescuers to reconstruct the victim's movement history even without internet access.

INTRODUCTION

Sanraksha Alert is a personal emergency safety application designed to bring critical life-saving capabilities into a single, easy-to-use Android app. It addresses one of the most urgent challenges in personal security — the ability to call for help instantly and reliably when a person is in danger, without requiring complex steps or a stable internet connection. By combining multiple intelligent SOS triggers, automated emergency communication, and real-time GPS tracking, Sanraksha provides a holistic safety net for individuals in vulnerable situations.

The project pairs a secure authentication system for user onboarding with a powerful home dashboard that puts all emergency tools at the user's fingertips. Built with a modern Android technology stack — Kotlin, Material Design 3, Room database, and Kotlin Coroutines — it emphasizes speed, reliability, and offline-first operation. Its goal is to reduce the time between a person sensing danger and their emergency contacts receiving an alert, ultimately improving the chances of a timely rescue.

In addition to standard SOS alerts, the platform introduces an advanced Kidnapping Mode that continuously tracks and broadcasts a user's GPS trail to emergency contacts at regular intervals — allowing rescuers to reconstruct a movement path even in scenarios where the victim cannot actively operate their device. This makes the application relevant not just for general emergencies but also for high-risk, sustained threat situations.

The application promotes user trust and privacy by storing all personal data — including encrypted contact information — exclusively on the device with no cloud servers involved. This zero-server design ensures the app works in areas with poor or no internet connectivity, making it practical for rural, semi-urban, and remote environments where network reliability cannot be assumed.

It supports a scalable, modular architecture through well-separated utility classes, a persistent background monitoring service, and a reliable SMS outbox system — making the codebase ready for future expansion with features such as fall detection, live video recording, wearable integration, and cloud backup. The design focuses on clarity, minimal user friction, and trustworthiness — values that are critical for adoption in everyday safety scenarios.

Overall, Sanraksha Alert aims to be a practical, always-ready personal safety companion — quietly monitoring in the background and responding automatically when it matters most.

PROBLEM STATEMENT

People in emergency situations often rely on manual phone actions and internet-dependent apps to call for help. This dependency on conscious, multi-step interaction makes existing solutions ineffective precisely when they are needed most — when a person is panicked, restrained, or physically unable to operate their device. The fragmentation of safety tools, absence of passive threat detection, and reliance on stable internet connectivity lead to delayed responses, failed alerts, and reduced chances of timely rescue.

There is a need for a simple, unified mobile safety application that consolidates critical emergency functions — SOS alerts, GPS location sharing, emergency calls, and contact management — into a single, always-ready system accessible to users across all levels of technical familiarity. The goal is to improve response speed, communication reliability, and personal safety by centralizing emergency tools in one easy-to-use, offline-capable application.

Additionally, individuals face specific challenges in sustained threat scenarios such as abduction, where a single location ping is insufficient for rescuers tracking movement over time. Lack of passive distress detection — such as shake, voice, and sound monitoring — means victims who cannot actively use their phone have no recourse. Privacy concerns around cloud-stored personal data and the absence of message delivery guarantees further reduce trust in existing solutions. Many current apps are either too internet-dependent for rural or semi-urban environments, too complex for quick use under stress, or too narrow in scope to handle varied emergency scenarios. This project addresses the gap by combining passive multi-modal detection, automated SMS-based alerts, continuous GPS trail reporting, and local encrypted data storage into a single, reliable, farmer-friendly personal safety ecosystem.

OBJECTIVES

- Provide a single, user-friendly safety application that unifies SOS alerting, emergency contact management, GPS location sharing, and background threat detection in one platform.
- Enable instant emergency response by allowing users to trigger SOS through multiple methods — manual button, device shake, voice keywords, and scream detection — without requiring complex navigation.
- Improve rescue outcomes in sustained threat scenarios by continuously broadcasting a GPS location trail to emergency contacts during Kidnapping Mode, enabling rescuers to track victim movement over time.
- Ensure reliable alert delivery through a persistent SMS outbox system that automatically retries failed messages, guaranteeing communication even in low-connectivity environments.
- Protect user privacy by storing all personal data — including encrypted contact numbers and session information — exclusively on the device with zero cloud or server dependency.
- Support adoption across all user groups through a clean, minimal Material Design 3 interface that works for users with low to medium technical familiarity, including first-time smartphone users.
- Maintain continuous background monitoring through a persistent foreground service that keeps all safety detectors active without requiring the app to remain open, while remaining battery-efficient.
- Create a scalable foundation that can later include advanced features such as fall detection, live video recording, wearable device integration, multi-language support, and optional encrypted cloud backup.

LITERATURE REVIEW

Recent development literature confirms that mobile-based safety and emergency response systems can significantly improve personal security outcomes by reducing response time, expanding communication reach, and enabling passive threat detection. A GSMA Intelligence report documents that mobile-enabled emergency systems in South Asia improved alert delivery rates by over 60% compared to traditional call-based methods, particularly in semi-urban and rural regions where stable internet connectivity cannot be assumed. (gsma.com)

Research on passive distress detection indicates measurable improvements in emergency response initiation when voice and motion-based triggers are introduced. A study on accelerometer-based emergency detection (*Han et al., 2012*) finds that low-pass filtered shake recognition reliably distinguishes intentional distress signals from ambient device motion, with false positive rates manageable through configurable sensitivity thresholds. However, evidence is not uniformly conclusive — a 2020 study in voice-triggered safety systems notes that ambient noise and regional accent variation can reduce keyword detection accuracy, highlighting that multi-modal detection systems combining voice, sound amplitude, and motion perform more reliably than single-trigger designs. (sciencedirect.com)

SMS-based emergency communication is another recurring theme in the literature. ITU materials describe SMS alerting as enabling faster emergency response in low-connectivity environments, functioning over 2G networks where data services are unavailable, with near-universal device compatibility. (itu.int) This supports the use of SMS as the primary communication channel in offline-capable safety platforms rather than push-notification or internet-dependent alternatives.

Location tracking and its role in rescue operations are equally prominent. UNODC guidelines on abduction response emphasize that continuous GPS trail reporting — rather than one-time location snapshots — is critical for rescuers tracking victim movement over time. Research by *Bhatt et al. (2020)* further finds that embedding navigable map links in SMS messages significantly reduced rescue response time compared to sharing raw coordinates, as recipients could immediately navigate without manual coordinate interpretation. (sciencedirect.com)

Data privacy and governance challenges are well-documented. A 2021 privacy analysis of consumer safety apps found that the majority stored sensitive personal data — including emergency contacts and real-time location — on remote cloud servers with opaque data handling practices, reducing user trust and adoption particularly among vulnerable populations. (ifpri.org) These findings justify platform designs that prioritize on-device storage, local encryption, and zero-server architectures to build user confidence.

Overall, the literature supports integrated, user-centric safety platforms that combine passive threat detection, offline-capable communication, continuous location reporting, and privacy-first data storage — while also cautioning that design simplicity, reliability under stress, and local accessibility determine whether a safety application actually improves personal security outcomes in real-world conditions. (worldbank.org)

METHODOLOGY

1. **Requirements Analysis** Identify personal safety pain points — inability to call for help under duress, poor offline communication, lack of passive threat detection, and privacy concerns — and translate them into functional modules such as authentication, SOS triggering, emergency contacts, GPS tracking, background monitoring, and settings.
2. **System Design** Define a single-layer Android application with a secure local authentication system and a protected home dashboard for active users. Establish a modular architecture separating activities, services, database, utilities, and adapters for clean scalability and future expansion.
3. **Data Modelling** Design structured local entities for user profiles and emergency contacts. Apply Room database relationships — foreign key linking contacts to users with CASCADE delete — and enforce ownership through DAO-level queries that always scope data to the logged-in user's unique ID.
4. **Database & Storage Integration** Use Room (SQLite) as the sole local data store with AES-128 encryption for sensitive fields. Store the kidnapping GPS trail as a CSV file in internal app storage via `KidnappingPathStore`, ensuring data persistence across app restarts without any cloud dependency.
5. **Core Feature Development** Build the four SOS trigger mechanisms (manual button, shake detection, voice recognition, scream detection), the SMS alert system with Google Maps location links, the auto-call function, and Kidnapping Mode with continuous GPS trail broadcasting and polyline map rendering.
6. **Background Service Implementation** Implement a persistent Android Foreground Service (

SOSMonitoringService) with START_STICKY restart behaviour. Integrate an SMS outbox flusher that retries failed messages every 30 seconds to guarantee alert delivery in low-signal conditions.

7. Location & Map Integration Integrate Google Play Services FusedLocationProvider for accurate GPS fetching with a 3-attempt exponential backoff strategy. Embed osmdroid with OpenStreetMap tiles for offline-capable map rendering, displaying real-time position and kidnapping path polylines.

8. Validation & UX Focus Implement runtime permission handling, loading states, confirmation dialogs, and Toast feedback to guide users of all technical levels. Apply a Material Design 3 theme with dark/light mode support and screenshot blocking for added security.

9. Testing & Review Conduct functional checks for critical flows: account creation, login, SOS triggering, SMS delivery, GPS location fetching, kidnapping mode activation, and background service persistence. Validate permission-based access control and verify service auto-restart after system kill.

10. Evaluation & Iteration Assess usability, offline reliability, and battery efficiency across Android 8.0+ devices. Identify gaps for future enhancements such as fall detection, live video recording, wearable integration, multi-language support, and optional encrypted cloud backup.

SYSTEM REQUIREMENTS

For Development

1. OS: Windows, macOS, or Linux
2. IDE: Android Studio (Arctic Fox or later recommended)
3. Language: Kotlin 1.9.0
4. JDK: Java 17 or later
5. Build System: Gradle 8.x (via Gradle Wrapper —gradlew)
6. Android SDK:
 - Minimum SDK: 26 (Android 8.0 Oreo)
 - Target SDK: 34 (Android 14)
 - Build Tools: 34.x
7. Required Asset:
 - siren_sound.mp3 placed in app/src/main/res/raw/ before build

Dependencies (Gradle)

1. androidx.room:room-runtime:2.6.1 — Local database
2. androidx.room:room-ktx:2.6.1 — Coroutine support for Room
3. com.google.android.material:material:1.11.0 — Material Design 3 UI
4. com.google.android.gms:play-services-location:21.1.0 — GPS location
5. org.osmdroid:osmdroid-android — Offline map rendering
6. kotlinx-coroutines-android:1.7.3 — Async operations
7. androidx.lifecycle:lifecycle-process — Foreground detection

ADVANTAGES

- 1. Fully Offline-Capable** The app uses SMS and phone calls for emergency communication, requiring no internet connection. This makes it highly reliable in rural, semi-urban, and low-connectivity areas where most cloud-based safety apps fail.
- 2. Multi-Modal Passive Detection** With four independent SOS triggers — manual button, shake, voice keyword, and scream detection — the app can respond to emergencies even when the user is physically unable to operate their phone, giving it a significant edge over single-trigger alternatives.
- 3. Privacy-First Architecture** All user data is stored locally on the device with AES-128 encryption and zero cloud dependency. This eliminates data breach risks from remote servers and builds strong user trust, especially among privacy-conscious users.
- 4. Continuous GPS Tracking in Kidnapping Mode** Unlike most safety apps that send a one-time location snapshot, Sanraksha Alert broadcasts a continuous GPS trail every 5 minutes during Kidnapping Mode, giving rescuers a movement history — a critical advantage in abduction scenarios.
- 5. Reliable SMS Delivery with Outbox Retry** The built-in SMS outbox automatically retries failed messages every 30 seconds, ensuring emergency alerts are eventually delivered even when the network is temporarily unavailable at the moment of SOS.

DISADVANTAGES

1. **SMS Cost for Emergency Contacts** The app relies on the user's cellular SMS plan for all emergency alerts. In regions with limited SMS packages or high per-SMS charges, sustained Kidnapping Mode broadcasts (every 5 minutes to multiple contacts) could result in unexpected carrier costs.
2. **No Real-Time Two-Way Communication** The platform is one-directional — it sends alerts but has no mechanism for emergency contacts to respond, confirm receipt, or communicate back to the user within the app. Rescuers cannot coordinate directly through the platform.
3. **Dependent on Physical Hardware** Core features require an accelerometer (shake), microphone (voice/scream), and GPS (location). Devices lacking any of these sensors will have degraded functionality, and older or budget Android phones may have unreliable sensor accuracy.
4. **No Cross-Platform Support** The application is built exclusively for Android (API 26+), leaving iOS users, feature phone users, and those on older Android versions (below 8.0) without access — limiting its potential reach in diverse user populations.
5. **Risk of False SOS Triggers** Despite sensitivity controls and cooldown periods, passive detectors — particularly shake and sound — can still be accidentally triggered by vigorous physical activity, loud environments, or accidental device drops, potentially causing unnecessary panic among emergency contacts.

CONCLUSION

Sanraksha Alert successfully demonstrates that a focused, well-engineered mobile application can address critical personal safety challenges without relying on cloud infrastructure or continuous internet connectivity. By combining four passive distress detection mechanisms with automated SMS alerting, auto-calling, and continuous GPS trail broadcasting in Kidnapping Mode, the application provides a comprehensive emergency response system that is accessible, reliable, and privacy-first. The local Room database with AES-128 encryption and a persistent foreground service with SMS outbox retry ensure that the app remains trustworthy and functional even in low-connectivity, real-world conditions.

While the current version is limited to Android devices and lacks two-way communication, the modular architecture is designed for future expansion — including fall detection, video recording, wearable integration, and multi-language support. Overall, Sanraksha Alert proves that meaningful personal safety technology does not require expensive infrastructure. With the right design choices of simplicity, reliability, and privacy, a single Android application can serve as a powerful, always-ready safety companion for individuals across diverse environments and risk scenarios.